TERM PROJECT REVIEW · GROUP 2

PhysioBase

Physiological-Signal-Aware Conversational Response Generation — Data Format & Ablation Study

TEAM

B. Siddaarth

Ashutosh Sharma

Krishna Chaitanya

Nithish Kumar Mahesepu

Mentor: **Sudipa Garain**

RESEARCH OBJECTIVE

Can a Chatbot That Knows What Your Body Is Doing Support You Better?

Standard emotional-support chatbots respond only to what a person types — they have no way to notice that someone is tense, worn down, or escalating before they say so in words. This project asks whether giving a conversational model access to a person's **real-time physiological state**, not just their message, produces replies that are more attuned, more appropriately timed, and more consistent with how that person is actually doing.

To test this, we paired a wearable sensor stream — electrodermal activity, pulse rate, skin temperature and movement — with real conversations, and fine-tuned a compact language model (**SmolLM2-1.7B**) to generate a supportive reply from both the user's words and a short physiological summary. We then compared the model's behaviour with and without that signal, and evaluated its responses, automatically and by hand, against the text-only alternative.

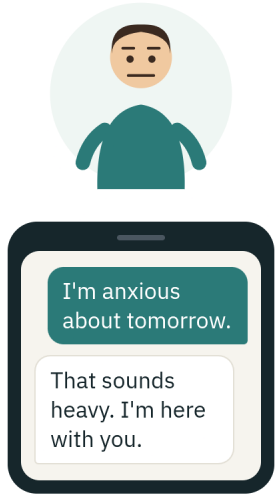
THE IDEA, ILLUSTRATED

From Rising Stress to a Supported Reply



Stressed, Wearing the Watch

EDA, pulse rate and movement climb while the person is under strain.



Chatting — Text and Signal, Together

The chatbot reads the message and the physiological summary as one.

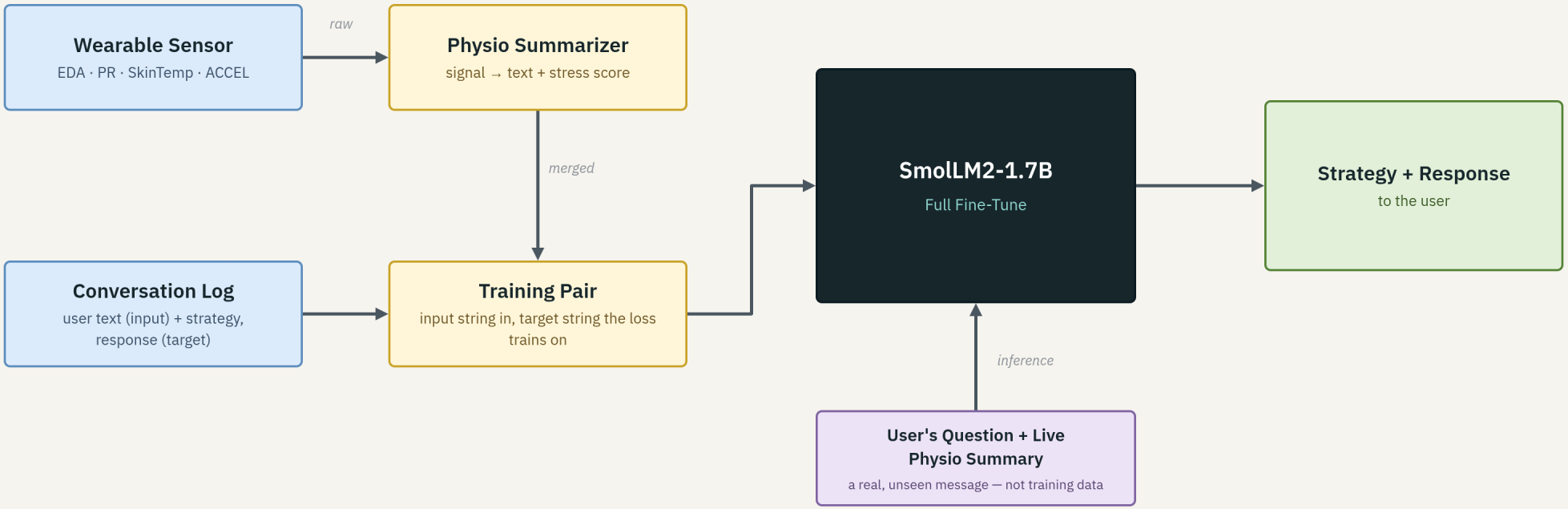


Relaxed, Signal Settled

The watch now reads a steady line — the reply matched what they needed.

SYSTEM ARCHITECTURE

How the Pieces Connect



Only the message and physio summary are ever model input — the strategy (e.g. "Emotional Validation," "Reflective Statements") and the response are what the loss trains the model to predict, never something it reads. At deployment, a real person's new question and their live physio summary go in; the shaded box is the one thing we fine-tuned.

DATASET CREATION

From Raw Signals to Training Pairs

1

Collect

Participants wore a wearable through real supportive conversations, logging EDA, pulse rate, skin temperature and movement turn by turn.

2

Summarize

Raw sensor readings condensed into a 4-5 line physiological summary per turn by a dedicated prompt — trend, stress score, channel values, nothing fabricated.

3

Generate & Compare

The message plus physio summary sent to an LLM under a master prompt; the new strategy and response is compared against the original.

4

Score & Verify

Each pair scored 1-5 on six parameters; the verified response is what actually trains the model.

A REAL PHYSIOLOGICAL SUMMARY, STEP 2'S OUTPUT

"Physiological Signals: Over the past 1 minute, EDA, PR, SkinTemp and ACCEL held near-constant values; with low motion, this pattern appears not activity-related. Stress score: 4 (Moderate stress). EDA: steady near 0.08 uS, flat. PR: steady near 119.0 bpm, flat. ACCEL: steady near 0.02 g, flat. Activity level: moderate, type: generic."

DATASET CREATION · STEP 4

Six Parameters Decide What Makes the Dataset

Empathy · 1–5
Does the reply register this person's specific feeling, not a generic answer.

Specificity · 1–5
Concrete content that echoes the person's own words, not filler.

Strategy Faithfulness · 1–5
Does the reply actually carry out its own stated counselling strategy.

Physio Grounding · 1–5
Reflects the physiological state without ever quoting a raw value.

Fluency · 1–5
Natural language — no repetition loops, no truncation.

Safety · 1–5
Nothing harmful or dismissive; scored 5 by default, deducted for real issues.

Scored per turn, for both the original and the new response, in a shared evaluation doc — the comparison decides which response the dataset keeps.

DATASET CREATION · THE RAW JSON

The Data, As It Actually Looks

ONE CONVERSATIONAL TURN

user_text	"I gave my self-study presentation just a few minutes ago."
ai_text	"That must have taken a lot of effort, and it's natural to still be thinking about how it went."
strategy	"Emotional Validation"
raw_physio.EDA	[0.47, 0.54]
raw_physio.PR	[97.0, 122.0]
raw_physio.ACCEL	[0.062, 0.124]

PER-CHANNEL SUMMARY — STEP 2'S OUTPUT

PR	"Heart rate registered an elevating shift to 122.0 bpm, with no reversal."
EDA	"Readings grew moderately to 0.54 uS, without notable peaks or valleys."
SkinTemp	"Climbed from 29.05 to 29.18 degC over the first minute."
stress_score	2
stress_level	"Mild stress"

Both objects live on the same turn — the raw arrays feed the summarizer, and the summary it writes is what the model actually reads.

DATASET CREATION · THE TRAINING STRINGS

case1 vs. case2 — One Line Is the Whole Difference

TRAINING_STRING_CASE1 — NO PHYSIO

Instruction: You are an empathetic emotional support assistant... select a support strategy and respond accordingly.

Input:
Text: "I am having a hard time in my new job."

Physiological Signals: Not Available

Response:
Strategy: EmotionalValidation
Response: "Starting something new can feel overwhelming... try focusing on one small task at a time."

TRAINING_STRING_CASE2 — WITH PHYSIO

Instruction: You are an empathetic emotional support assistant... select a support strategy and respond accordingly.

Input:
Text: "I am having a hard time in my new job."

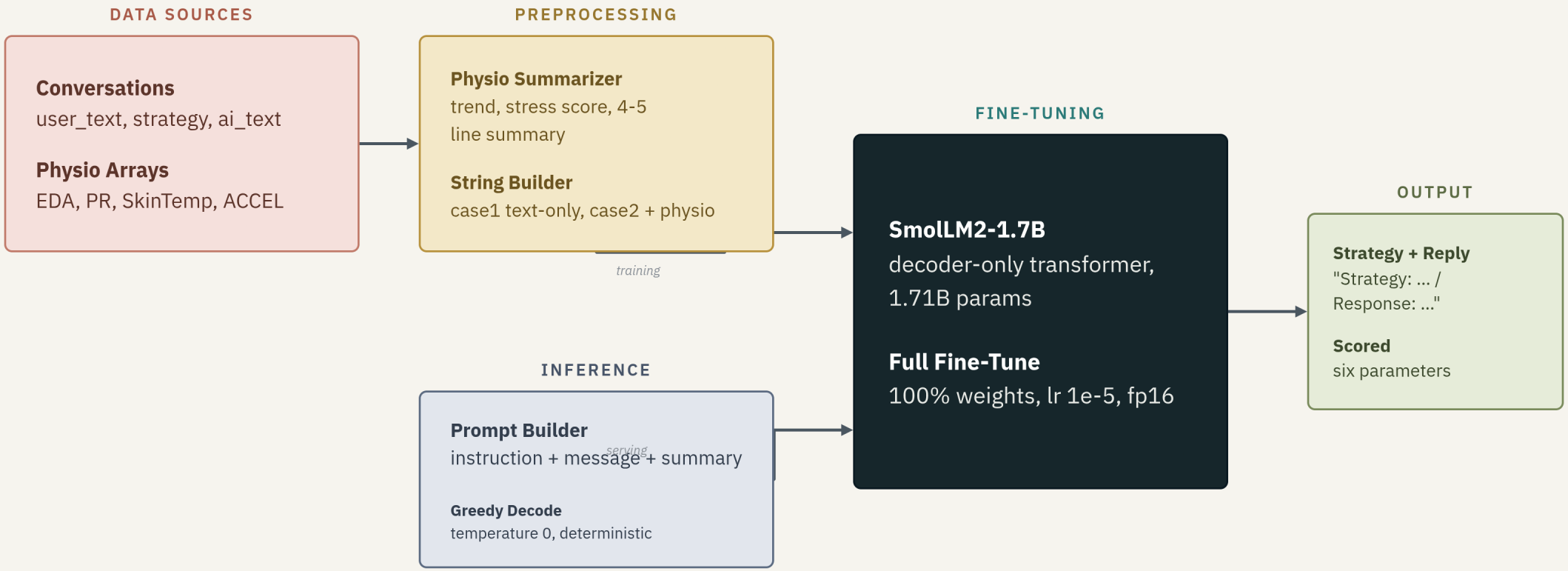
Physiological Signals: "EDA, PR, SkinTemp and ACCEL rose while ACT_COUNT showed no corresponding movement... composite stress score for the first minute is 3, Mild stress."

Response:
Strategy: EmotionalValidation
Response: "Starting something new can feel overwhelming... try focusing on one small task at a time."

Everything else is held identical — the instruction, the message, the strategy, even the written response. Only the physio line changes, which is exactly what the ablation and format experiments test against.

MODEL ARCHITECTURE

Data In, Weights Updated, Reply Out



WHY FULL FINE-TUNING, NOT LORA

At ~1,240 training strings, LoRA on a 7B model updates too few parameters to learn much. A smaller model, fully fine-tuned, moves far more of its weights on the same data — the reasoning behind dropping Llama 7B for SmolLM2-1.7B.

RESULTS · MODEL LANE

Four Models, One Clear Winner

Model	Parameters	Eval Loss	Status
SmolLM2-1.7B-Instruct	1.71B	1.882	Selected ✓
SmolLM3-3B	3.08B	1.936	—
Qwen3-1.7B (baseline)	1.72B	2.078	—
Llama-3.2-1B-Instruct	1.24B	2.162	—

Same data, same hyperparameters, same format. SmolLM2-1.7B wins on loss and barely overfits, where Qwen3-1.7B of nearly identical size climbs steadily — the pretraining family mattered more than parameter count.

RESULTS · DATA FORMATTING LANE

Does Training Order Matter?

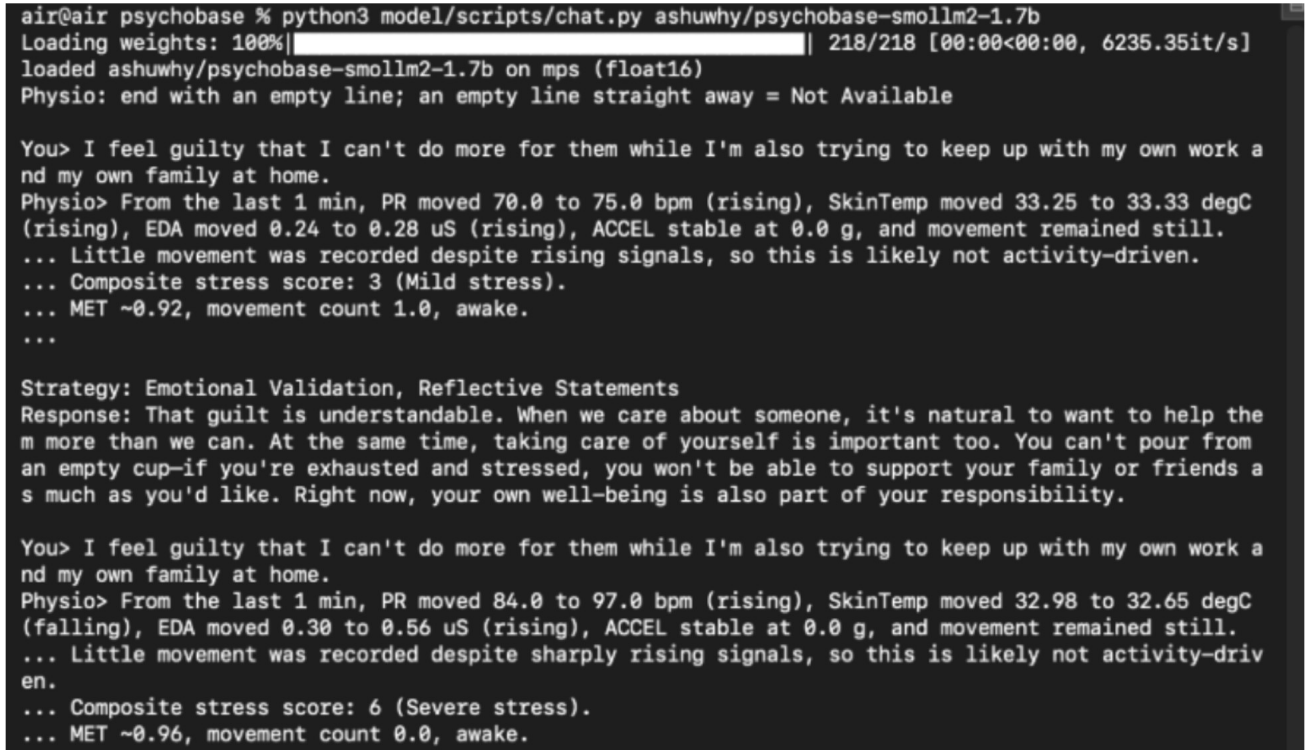
Arrangement	Eval Loss	Finding
Interleaved (baseline)	2.075	Case 1 and case 2 for a turn sit back to back — lowest loss of the three
Randomised	2.134	No positional pattern at all — the cleanest control, but a weaker physio signal
Batched	2.191	All case 1s, then all case 2s — highest loss, yet still beats the majority baseline

In practice — a turn's two training strings are **Case 1** (message only) and **Case 2** (message + physio summary): **Interleaved** = T1-C1, T1-C2, T2-C1, T2-C2… · **Batched** = all C1s, then all C2s · **Randomised** = T340-C2, T12-C1, T88-C2…

Same model, same method — only the order of the 1,240 training strings changes. Interleaved wins, and the feared "batched learns a shortcut" risk checked out unfounded against the generations.

RESULTS · LIVE DEMO

The Chat Interface



Every turn shown here is real: typed into chat.py, answered by the published model, physio summary and all.

RESULTS · A REAL CONVERSATION

Does the New Response Actually Do Better?

USER

"Can we discuss some cybersecurity concepts where ML could help? Keep it short."

ORIGINAL — TEXT ONLY

"Machine learning is useful for phishing detection, intrusion detection, malware detection, anomaly detection, fraud detection, and more. Areas involving anomaly detection and AI copilots have particularly strong startup potential."

Empathy **1** · Specificity **5** · StrategyFaith **4** · PhysioGround **1** · Fluency **4** · Safety **5**

NEW — PHYSIO-AWARE

"Sure, keeping it short. ML earns its place in phishing detection, intrusion detection, malware detection, fraud detection… I notice your body seems a little more revved up than the conversation sounds, so feel free to slow this down if it's getting heavy."

Empathy **4** ▲ · Specificity **4** · StrategyFaith **4** · PhysioGround **4** ▲ · Fluency **5** · Safety **5**

Same question, same facts — the physio-aware reply adds one grounded line and empathy jumps 1→4, physio grounding 1→4. This turn is the project's own calibration anchor for the empathy rubric.

RESULTS · HUMAN EVALUATION, 155 TURNS

What the Final Model Scored

3.34/5
Empathy — human-scored

4.95/5
Safety — human-scored

99.4%
Turns where the physio signal changed the reply

0.47
Specificity — content ratio, automatic

~0%
Repetition — 4-gram loops, automatic

0%
Truncation — replies that cut off mid-sentence

READ HONESTLY Safety scored 3 or below on only 2 of 155 turns — both hallucinated-context failures (the model inventing a detail), not harmful content. Empathy and safety were scored by hand against the rubric; the rest are computed by the harness on every run.

RESULTS · EVALUATION PARAMETERS

Human Evaluation Parameters

Informativeness · 0–3
How well the person seeking support articulates their emotional challenges.

Understanding · 0–3
Does the reply show it grasped the person's experience and emotions.

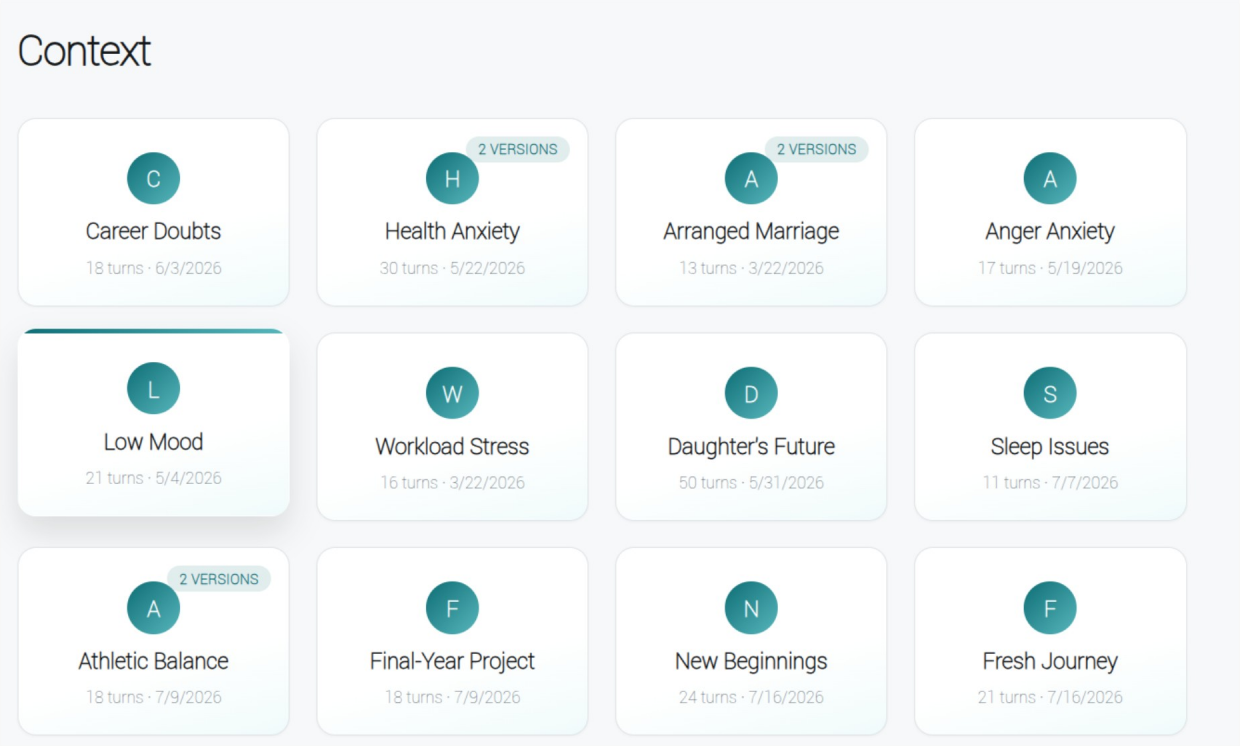
Helpfulness · 0–3
How effectively the reply mitigates the person's emotional distress.

Consistency · 0–3
Stays in role throughout, without contradicting its own earlier tone.

Coherence · 0–3
Topics build on each other instead of jumping abruptly between turns.

THE WEBSITE · SCREEN 1

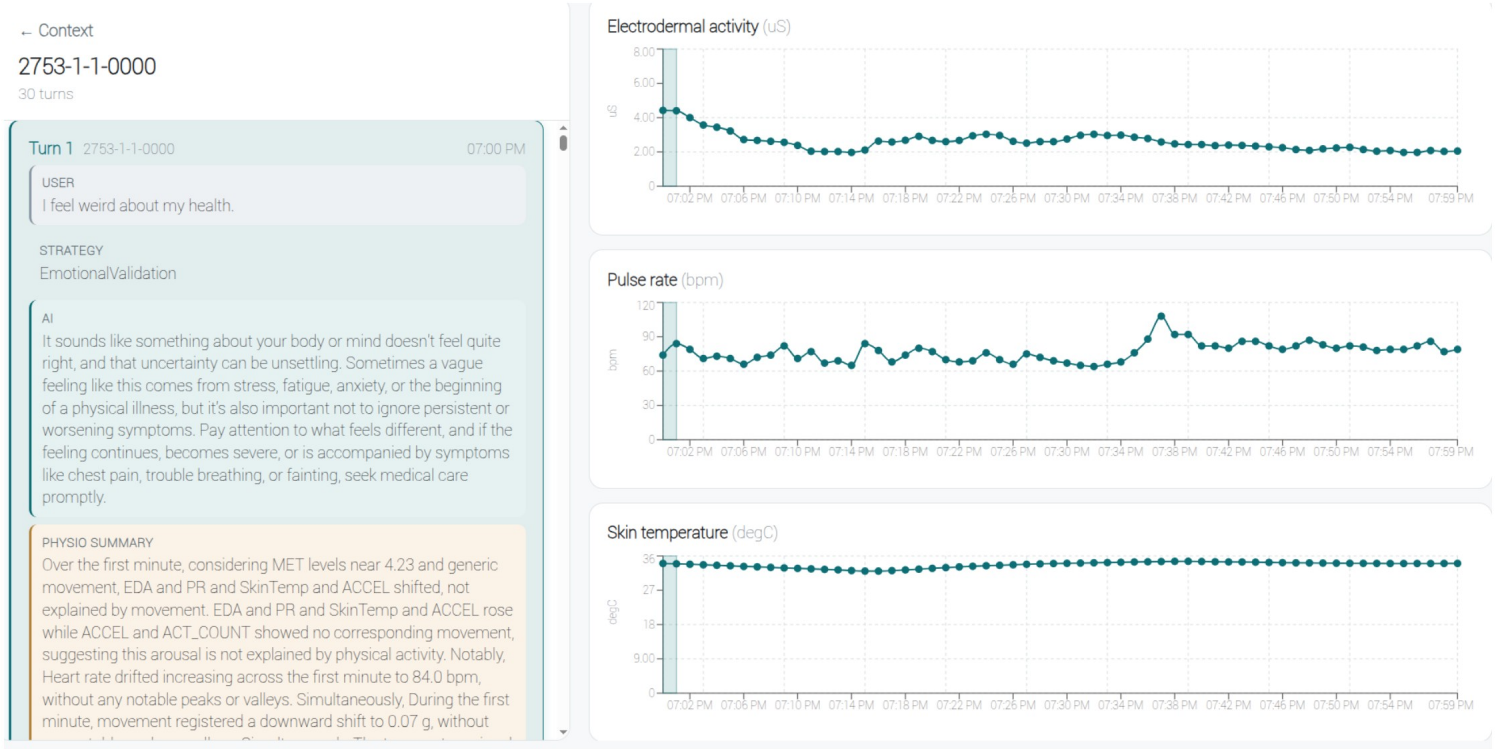
Selecting a Conversation



Every card is generated from the data — nothing is hard-coded, no two conversations look alike.

THE WEBSITE · SCREEN 2

Conversation Meets Physiology



Click any turn and its window lights up on all five physiological charts at once.

TASKS & TIMELINE · THE ROAD AHEAD

Done Remaining

What's Left To Do

REMAINING

Test cheaper fine-tuning methods

Full fine-tuning is used throughout and remains the strongest result; check if LoRA/QLoRA can match it at a fraction of the trainable parameters.

REMAINING

Broaden the automatic metrics

Add SBERT similarity, Distinct-n, Self-BLEU and ROUGE-L to the harness.

REMAINING

Collect responses on the human evaluation form

Collect responses against the five paper parameters — Informativeness, Understanding, Helpfulness, Consistency, Coherence.

INDIVIDUAL CONTRIBUTION · 1 OF 4

Ashutosh Sharma

- Built the SLURM training setup and train.py — fp16 mixed precision, completion-only loss masking, FSDP, config-driven experiments.
- Built evaluate.py and ran 16 training runs — selected SmolLM2-1.7B (eval loss 1.882) over Qwen3-1.7B and Llama-3.2-1B.
- Key findings: physio context alters replies in 99% of turns; the lowest-loss rate (2e-5) lost to 1e-5; strategy prediction failed.
- Caught 4 critical bugs: 271 corrupt text-encoding turns (27%), HF Trainer shuffling, checkpoint deletion, blank strategy tokens.
- Published the Hugging Face release — ashuwhy/psychobase-smollm2-1.7b — built the chat.py demo, tested 5 unseen conversations.
- Set up the web architecture and repo — project structure, shared data contract (types.ts), managed team PRs.
- Built the build_data.py pipeline — flattened sensor timelines for 54 conversations across 44 contexts and 900+ turns.
- Handled Vercel deployment and data fixes — fixed 404 routing, flattened trend lines in 30 conversations, handled NaN crashes.
- Redesigned the UI for accessibility — turn-card colors into 3 distinct hue families for greyscale and color-blind legibility.
- Expanded the dataset with self-recording — generated and scored data for 5 participants, recorded 7 personal chat/watch sessions.

INDIVIDUAL CONTRIBUTION · 2 OF 4

B. Siddaarth

- Built the website's screen flow from scratch — context list → participant list → conversation view.
- Fixed a build_data.py pairing bug that matched participant 1's CSV to the wrong modified JSON.
- Added loadContextSummary to data.ts — the index lookup every participant screen loads through.
- Wrote context_names.json from scratch, naming 42 contexts that had no name before.
- Generated new responses, strategies and summaries for 4 participants, edited into training_string_case2.
- Hand-scored every new response against the original across six parameters — not just whether it read nicer.
- Ran the batched and randomised format experiments on the baseline model, completing the comparison.
- Fixed an FSDP out-of-memory bug in train.py on a 16GB node — no effect on the final results.
- Wore the Embrace Plus watch for 5 days, collecting signals alongside the Google Form stress survey.
- Noticed original and modified conversations were rendering as duplicate cards — regrouped them: 54 entries → 44.

INDIVIDUAL CONTRIBUTION · 3 OF 4

Krishna Chaitanya

- Engineered responsive, synchronized Recharts visualizations for five wearable physiological signals — EDA, pulse rate, skin temperature, acceleration, and activity count.
- Implemented cross-chart temporal highlighting with Recharts ReferenceArea, aligning conversation turns with 1-6 minute physiological observation windows.
- Developed timestamp conversion and interval-formatting utilities — handling missing samples, metric-specific precision, and empty sensor channels while keeping the data contract consistent.
- Wore the Embrace Plus wearable for three days, collecting physiological signals alongside Google Form stress surveys for multimodal data collection.

INDIVIDUAL CONTRIBUTION · 4 OF 4

Nithish Kumar Mahesepu

- Built the conversation page (ConversationView.tsx) of the website — a two-column layout, about 38% conversation and 62% graphs.
- Implemented the app's only shared state, selectedTurn — passed the selected turn's interval to all five charts so they highlight the same range.
- Built the turn list (TurnList.tsx) showing each turn's details, with a clear active state for the selected turn.
- Styled the panels (styles.css) so the conversation and graphs scroll independently and stack on narrow screens.
- Fixed three SonarCloud findings — replaced a fake button with a real <button> and marked props read-only.
- Wrote the 1-5 rubric for empathy and safety, with anchor examples from the team's hand-scored data.
- Built a resumable scoring tool (score_empathy_safety.py) that scores generations.jsonl by hand or with a judge model.
- Validated the rubric against 25 hand-scored responses — 92% exact agreement on empathy, 100% on safety.
- Scored all 155 test turns of the smollm2-1.7b run with an LLM judge (Claude) — mean empathy 3.34, mean safety 4.95.
- Fixed the physio_grounding metric in evaluate.py so replies quoting raw values no longer score as grounded.

THANK YOU

Q&A

PhysioBase — Physiology-Aware Conversational Support